

1st Tutorial examination
Department of Chemistry
Course No-322
Marks -10, Time: 40 min.

1. Describe the Band Theory of Solids 4
2. Explain (i) Conductor (ii) Insulator (ii) Semi- conductor in terms of Band Theory 3
3. Define: (i) Fermi Level (ii) Doping (iii) Solid Solution. 3

Summon-47

Answer – 47

1st Tutorial examin. on
Department of Chen stry
Course No-332
Marks -20, Time: 4(nin.

1. (a) Describe the Band Theory of Solids
(b) Explain (i) Conductor (ii) Insulator (ii) Semi- cond tor in terms of Band Theory 4 6

2. (a) What is borazine? Justify borazine as an electronic omer of benzene 1+2

(b) Draw the structures of B_5H_9 and $B_{10}H_{14}$ and discus: he unusual bonding in diborane. 2+2+3

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solid. Every m
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2nd Tutorial examination
Department of Chemistry
Course No-332
Marks -20, Time: 40 min

1. (a) Discuss the properties of Ionizing Solvent. 2
(b) Compare the properties of liquid NH_3 and H_2O. 4
(c) Write the self ionization reactions of liquid SO_2 and liquid NH_3 with example. 4

2. (a) How are $[NPCl_2]_3$ and $[NPCl_2)_4$ produced ? 2
(b) Discuss the salient feature of structure and bonding in $[NPCl_2]_3$. 4
(c) Synthesis S_4N_4. How does (SN)x behave like a one dimensional conductor? 4

Figure 3: Sino- India Nuclear Capabilities

Department of Chemistry, JU
1st Tutorial Examination 2017
Course No. Chem. 322H, Time: 30 minutes

1. Define solvolysis and solvation reaction with suitable example.
2. Explain why solution of alkali metal in liq. NH_3 is blue colored.
3. In liquid ammonia solution the strong oxidizing agent aren't stable. why?
4. Complete the following reactions (any four)

(i) $KI + NH_4Cl \xrightarrow{\text{Liq. } NH_3}$

(ii) $BF_3 + NH_3 \xrightarrow{\text{Liq. } NH_3}$

(iii) $PCl_5 + SO_2 \xrightarrow{\text{Liq. } SO_2}$

(iv) $PH_3 + Na \xrightarrow{\text{Liq. } NH_3}$

(v) $2KI + SOCl_2 \xrightarrow{\text{Liq. } SO_2}$

2nd Tutorial Examination 2017
Department of Chemistry
Course # Chem. 322
Marks: 10, Time: 40 min.

1. (a) What are metals? Mention some characteristic properties of metals. 1.25
(b) Distinguish among conductors, semiconductors and insulators on the basis of band theory of solids. 3
(c) As the temperature increases the electrical resistance of metal increases – explain. 2
(d) What is called doping. What are the p and n - type semiconductors? 2.75
(e) What is called super lattice? 1

2. (a) What is silicone polymer? Discuss the preparation of silicons. 1+2=3
(b) How phosphazene polymer obtained? 2
(c) What happens when $[NPCl_2]_3$ (i) hydrolyzed with water and (ii) reacts with excess of NH_3? 1+1=2
(d) Explain the structure of S_4N_4 compound. 3

2. (a) Discuss the main feature of Band Theory of metals. 4

(b) Show the difference among conductors, semiconductors and insulators on the basis of band theory of solids. 3

(c) What is called Fermi Level? What are the P and N-type semi conductors? 2+2

(d) As the temperature increases the electrical resistance of metal increases-explain. 3

3. (a) What is Silicone polymer? Discuss the preparation of Silicones. 1+2

(b) How are $[NPCl_2]_3$ and $[NPCl_2]_4$ produced. Discuss the salient feature of structure and bonding in $[NPCl_2]_3$ 2+4

(c) What happens when $[NPCl_2]_3$ (i) is hydrolyzed with water, (ii) reacts with excess of NH_3, (iii) reacts with NaF and (iv) $NaOCH_3$.

(d) Silicone and phosphazene are isoelectronic-explain

Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Examination 2018
Course Title: Inorganic Chemistry IV (Selected Topics)
Course: Chem. 322
Full Marks: 70, Time: 4 hrs.

Answer any FIVE questions

1. (a) Explain the following terms with example: 4
   (i) Plane of symmetry,
   (ii) Center of symmetry,
   (iii) Improper axis of rotation and
   (iv) Symmetry operation
   (b) Draw sketches to indentify the following symmetry elements- 4
   (i) A σ_v and a σ_h in BF_3 (ii) A C_3 axis and a σ_v plane in $CHCl_3$
   (c) Use a flow chart to assign the point group to each of the following. 6
   (i) BF_3, (ii) $CHCl_3$ and (iii) B_2H_6

2. (a) Discuss the main feature of Band Theory of metals. 4
   (b) Show the difference among conductors, semiconductors and insulators on the basis of band theory of solids. 3
   (c) What is called Fermi Level? What are the P and N-type semi conductors? 2+2
   (d) As the temperature increases the electrical resistance of metal increases-explain. 3

3. (a) What is Silicone polymer? Discuss the preparation of Silicones. 1+2
   (b) How are $[NPCl_2]_3$ and $[NPCl_2]_4$ produced. Discuss the salient feature of structure and bonding in $[NPCl_2]_3$ 2+4
   (c) What happens when $[NPCl_2]_3$ (i) is hydrolyzed with water, (ii) reacts with excess of NH_3, (iii) reacts with NaF and (iv) $NaOCH_3$. 2
   (d) Silicone and phosphazene are isoelectronic-explain 3

4. (a) What do you mean by essential and trace elements in biological process? 2
   (b) What are meant by porphine and metalloporphyrins? 2
   (c) Explain Deoxy-Mb is paramaganetic whearas Oxy-Mb is diamagnetic. 3
   (d) Hemoglobin acts as oxygen carrier whereas cytochrome acts as electron carrier although the active site for both of them is same. Explain why? 3
   (e) Show the oxygen saturation curves for myoglobin and hemoglobin against oxygen pressure. 4

5. (a) Discuss the methods for the separation of lanthanides from Monazite. 3
   (b) Explain why the 4f electrons of the lanthanide elements do not take part in chemical bonding. 2
   (c) Why La^{3+}, Ce^{4+} and Lu^{3+} are diamagnetic but all other lanthanides ions are paramagnetic-explain? 2
   (d) What are the differences and similarities between the lanthanides and actinides? 2
   (e) Actinides show higher oxidation state than lanthanides-explain? 2
   (f) Why are Eu^{2+} and Yb^{2+} more stable than those of Eu^{3+} and Yb^{3+} ions? 2
   (g) Why $Ce(OH)_3$ is more basic than $Lu(OH)_3$? 1

6. (a) What are the features of a substance to be a solvent? 3
(b) The solution of alkali metals in liquid ammonia is paramagnetic and the color of the solution is blue – explain.
(c) Explain the following with proper chemical equation: 2
(i) Solvolysis of Cl_2 in NH_3 is parallel to that in H_2O. 3
(ii) CH_3COOH a weak acid in H_2O but behaves as a strong acid in NH_3.
(iii) Urea shows no acidic behavior at all in H_2O but behaves as a strong acid in NH_3.
(d) Compare the properties of liquid NH_3 and H_2O as ionizing solvents. 2
(e) Complete the following reactions (any four): 4

(i) $PCl_5 + SO_2 \xrightarrow{\text{Liq. } SO_2}$

(ii) $FeCl_3 + KI \xrightarrow{\text{Liq. } SO_2}$ $FeCl_2 + KCl + I_2$

(iii) $HClO_4 + H_2SO_4 \xrightarrow{\text{Liq. } H_2SO_4}$

(iv) $CH_3COOH + H_2SO_4 \xrightarrow{\text{Liq. } H_2SO_4}$

(v) $NH_2CONH_2 + H_2SO_4 \xrightarrow{\text{Liq. } H_2SO_4}$

7. (a) Draw the structures of B_5H_9 and $B_{10}H_{14}$. Show by orbital diagram the formation of 3c-2e open and closed BBB bonds.
(b) Why borazine is called an inorganic benzene? 2+3
(c) What happens when diborane reacts with (i) excess ammonia at low temperature and (ii) water? 3
(d) "Boron hydrides are high potential hydrogen storage materials"-Justify the statement. 2 4

8. (a) Define the following terms: (i) Diamagnetism (ii) Ferromagnetism (iii) Antiferromagnetism 3
(b) What do you understand by temperature independent magnetism? 2
(c) What is Curie point? Explain the effect of temperature on susceptibility of ferromagnetic substance
(d) What is magnetic moment? Which of the following nuclei should have zero magnetic moment: 5
$^{2}_{1}H$, $^{12}_{6}C$, $^{14}_{7}N$, and $^{16}_{8}O$ 4

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part- III Final Examination 2017
Course # Chem. 322
Full Marks: 35, Time: 2.5 hrs.

[Answer any Four questions]

1. (a) Mention three prime properties of metal. 0.75
(b) What are the major crystal structures observed in metals? 0.50
(c) (i) Define Heats of Atomization (ii) Explain the fact that Melting Point decreases Across the group Li-Na-K-Rb-Cs. 0.5+1
(d) Explain the MOT in Li_4 molecule. 2
(e) (i) Define alloys. 0.5
(ii) Give Hume-Rothery Rules those define the miscibility of metals in alloys. 1.25
(iii) Explain the fact that Li/Na or K/Na alloys don't exist. 1
(f) What is a Perovskite structure? Give the formula and draw the structure of a common Perovskite material. 1.25

2. (a) What are non-aqueous solvents? Name three non-aqueous solvents. 0.5+1
(b) Addition of Alkali metal to NH_3 creates an intense blue solution with electrical conductivity comparable to metal. Explain the following: 1+1+1
(i) 'Solvated electron' theory.
(ii) The blue solution is paramagnetic.
(iii) The blue solution is a good one-electron reducing agent.
(c) Explain the following with proper chemical equation: 0.75×3
(i) Solvolysis of Cl_2 in NH_3 is parallel to that in H_2O.
(ii) CH_3COOH, a weak acid in H_2O but behaves as a strong acid in NH_3.
(iii) Urea shows no acidic behaviour at all in H_2O, but behaves as weak acid in NH_3.
(d) Complete the following reactions (any four): 0.5×4=2

(i) $PCl_5 + SO_2 \xrightarrow{\text{liq. } SO_2}$

(ii) $FeCl_3 + KI \xrightarrow{\text{liq. } SO_2}$

(iii) $HClO_4 + H_2SO_4 \xrightarrow{\text{liq. } H_2SO_4}$

(iv) $CH_3COOH + H_2SO_4 \xrightarrow{\text{liq. } H_2SO_4}$

(v) $NH_2CONH_2 + H_2SO_4 \xrightarrow{\text{liq. } H_2SO_4}$

3. (a) What is an electron deficient compounds? Show by orbital diagram the formation of 3c-2e open and closed BBB bonds. 4
(b) Justify the borazine as an electronic isomer of benzene. 2.75
(c) What happens when diborane reacts with (i) excess ammonia at low temperature and (ii) water? 1+1

4. (a) What is silicone polymer? Discuss the preparation of silicons. 1+2
(b) How phosphazene polymer obtained? 2
(c) What happens when $[NPCl_2]_3$ (i) hydrolyzed with water and (ii) reacts with excess of NH_3? 1+1=2
(d) Explain the structure of S_4N_4 compound. 1.75

5. (a) Explain the structural similarities and differences between haemoglobin in oxy-deoxy states. 2
(b) What are distal and proximal histidines? What are their role in oxygen transport? 2
(c) Draw structure of chlorophyll. 1.75
(d) What are cytochromes? Structurally how do they differ from haemoglobin? 2
(e) What is an apoprotein? 1

6. (a) What are lanthanides? Write down their common electronic configuration. 1.25
(b) What is Lanthanide contraction? State two consequences of lanthanide contraction. 2
(c) Name the various methods for extraction of lanthanides and discuss any one method for their extraction and separation. 3
(d) Explain the facts: 2×0.75
(i) Ln^{3+} ions do not form complexes readily like transition metals.
(ii) Ln^{3+} ions forms stable complexes with oxygen rich chelating ligands.

7. Write short note (any three) of the following: 3+3+2.75
(a) Metal carbides
(b) Fluorocarbons
(c) Nitrogen fixation
(d) Chemistry of fullerenes
(e) Vitamin B and B_{12} co-enzymes.

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Department of Chemistry
Jahangirnagar University
B. Sc. (Part III) Honours Examination 2016
Course Title: Inorganic Chemistry IV (Selected Topics)
Course No: Chem - 322
Time: 2.5 Hours, Total Marks: 35

Answer any four questions

1. (a) What are metals? Mention some characteristic properties of metals. 2
(b) Explain the band theory of metals. 0.75+2
(c) Based on band theory of metals clearly explain the following: 4
(i) Fermi level (ii) forbidden zone (iii) insulator and (iv) semiconductor

2. (a) What are the properties of solvent? How are solvents classified? 1+1
(b) Addition of alkaline-earth metal in liq. ammonia produces a complex, whereas the solution of alkali metal in liq. ammonia is coloured-explain. 2
(c) Compare the properties of liquid NH_3 and H_2O as ionizing solvents. 2.75
(d) Complete the following reactions (any four):. 0.5×4=2

(i) $KI + NH_4Cl \xrightarrow{\text{liq. } NH_3}$

(ii) $NH_4Cl + KNH_2 \xrightarrow{\text{liq. } NH_3}$

(iii) $BCl_3 + NH_3 \xrightarrow{\text{liq. } NH_3}$

(iv) $PCl_5 + SO_2 \xrightarrow{\text{liq. } SO_2}$

(v) $FeCl_3 + KI \xrightarrow{\text{liq. } SO_2}$

(vi) $SbCl_3 + KCl \xrightarrow{\text{liq. } SO_2}$

3. (a) What is Lanthanide contraction? What are the consequences of Lanthanide contraction? 2
(b) Name the various methods for the extraction of Lanthanide and discuss one method for the extraction and separation of Lanthanide. 3
(c) State the colour of actinide ions. 1.25
(d) What are the similarities and differences between Lanthanide and Actinide elements? 2.5

4. (a) Explain the similarities and differences in structures between myoglobin and hemoglobin in the deoxy- and oxy- states. 2
(b) Why are the distal and proximal histidines in myoglobin or hemoglobin? 1
(c) What is oxyhemoglobin dissociation curve? Draw the standard oxyhemoglobin Dissociation Curve and explain the factors that govern the dissociation curve. 1+2+2
(d) What are the structural features of Chlorophyll? 0.75

5. (a) What is called polymer? How is polymer classified? 1.75
(b) How are $[NPCl_2]_3$ and $[NPCl_2]_4$ produced? 2
(c) Discuss the salient feature of structure and bonding in $[NPCl_2]_3$. 3
(d) Draw the formulae of one polymer having Si-O, B-N and S-S bond. 2

6. (a) What are intermetallic compounds? Mention their properties 2.5
(b) Liquid SO_2 solutions of both water and hydrogen bromide are nonconductors of electricity but a mixture of the two becomes a good conductor-Explain why. 1.75
(c) What are borohydrides? Draw the structure of the compounds (i) $Be(BH_4)_2$, (ii) $Al(BH_4)_3$. 2
(d) Write a brief note on fluorocarbons. 2.5

7. (a) Write short notes on (**any three**) of the following: 3+3+2.75
(a) Metal carbide
(b) Vitamin B_{12}
(c) Solar Cell
(d) Hydrides of Borane and their usage as reducing agents
(e) Fullerene-C_{60}

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Department of Chemistry
Jahangirnagar University
B. Sc. (Part III) Honours Examination 2015
Course Title: Inorganic Chemistry (IV) (Selected Topics)
Course No.: Chem - 322
Time: 2.5 Hours, Total Marks: 35

Answer any four questions

1. (a) Distinguish among conductors, semiconductors and insulators on the basis of band theory of solids. 3.0
   (b) As the temperature increases the electrical resistance of a metal increases-explain. 2.0
   (c) What is called doping? What are the p and n-type semiconductors? 2.75
   (d) What is called supper lattice? 1.0

2. (a) Explain why the 4f electrons of the lanthanide elements do not take part in chemical bonding. 1.5
   (b) Discuss the methods for the separation of lanthanides from monazite. 4.0
   (c) Discuss the colour of actinide ions. 1.5
   (d) Actinide show higher oxidation state than that of lanthanide-explain. 1.75

3. (a) Hemoglobin acts as oxygen carrier whereas cytochrome acts as electron carrier although the active site for both of them are same. Explain why? 2.0
   (b) Show the oxygen saturation curves for myoglobin and hemoglobin against oxygen pressure. What are the factors that govern the saturation curves? 2+1 = 3
   (c) Draw the structure of vitamin B_{12}. 2.0
   (d) Sketch the block diagram of 'Nitrogen Cycle." 1.75

4. (a) What is an electron deficient compound? Show by the orbital diagram the formation of 3C-2e open and closed BBB bonds. 2+ 2 =4
   (b) Justify the borazine as an electronic isomer of benzene. 3.0
   (c) Explain the order of acidity for the following series. 1.75
   $BF_3 < BCl_3 < BBr_3$

5. (a) Draw the structures of B_5H_9 and $B_{10}H_{14}$ and discuss the unusual bonding in diborane. 1+1=2
   (b) What happens when $[NPCl_2]_3$ is heated to 250°C and followed by treatment with (i) RONa and (ii) RNH_2 2.0
   (c) Explain why boric acid is a weak monobasic acid but behaves as a strong acid in Presence of a 1,2-diol compound. 2.5
   (d) "B-N bond in borazine is slightly polar but C-C bond in benzene is non-polar"-explain. 2.25

6. (a) What are the features of a substance to be a solvent? 1.0
   (b) How are the solvents classified? 2.0
   (c) The solution of alkali metals in liquid is paramagnetic and the colour of the solution is blue-explain. 2.0
   (d) Acetic acid is a weak acid in H_2O solution but behaves as a strong acid in liquid NH_3 - explain. 2.0
   (e) Separate solutions of liquid SO_2 in H_2O and liquid SO_2 in HBr cannot conduct electricity but mixture of the two solutions is a good conductor of electricity – explain. 1.75

7. (a) What are metalloporphyrins? Give one biological importance of metalloporphyrin. 2.0
   (b) What is photosynthesis? What is the function of chlorophyll molecule in photosynthesis. 2.0
   (c) What happens when biological systems containing hemoglobin are exposed to CO? 2.0
   (d) Draw the structure of porphyrin and explain how the structures of heme and chlorophyll are related to it? 2.75

Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Examination 2020
Course Title: Inorganic Chemistry IV (Selected Topics)
Course # Chem. 322, Full Marks: 35, Time: 2.5 hrs.

[Answer any five questions]

1. (a) Mention some characteristic properties of metals. 1.0
   (b) Show the difference among conductors, semiconductors and insulators on the basis of band theory of solids. 3.0
   (c) What is called Fermi Level? What are the P and N-type semi conductors? 3.0

2. (a) Discuss the properties of Ionizing solvent. 2.0
   (b) Compare the solvent properties H_2O and liquid NH_3. 2.0
   (c) The dissolution of alkali metal in liq. NH_3 produce a
   (i) Blue Coloration (ii) is a good conductor of electricity – explain. 1.5+1.5

3. (a) What is lanthanide contraction? What is the consequence of lanthanide contraction? 1+1
   (b) Discuss the ion-exchange method for the separation of lanthanides. 2.0
   (c) What are the points of differences and similarities between the lanthanides and actinides? 1.5
   (d) How do the lanthanide complexes act as NMR shift reagent? 1.5

4. (a) Hemoglobin acts as oxygen carrier whereas cytochrome acts as electron carrier although the active site for both of them are same. Explain why? 2.0
   (b) Show the oxygen saturation curves, for myoglobin and hemoglobin against oxygen pressure. 3.5
   (c) What is the Function of Chlorophyll? 1.5

5. (a) What is called polymer? How is polymer classified? 1.5
   (b) How are $[NPCl_2]_3$ and $[NPCl_2]_4$ produced? 2.0
   (c) Discuss the salient feature of structure and bonding in $[NPCl_2]_3$. 2.0
   (d) Draw the formulae of one polymer having Si-O, B-N and S-S bond. 1.5

6. (a) Draw the structures of B_5H_9 and $B_{10}H_{14}$. 2.0
   (b) Discuss the Structure / Bonding element of B_2H_6. 3.0
   (c) How is borazine prepared? Borazine is Inorganic benzene – explain. 2.0

7. Write short notes (any three) of the following: 2.5+2.5+2
   (a) Metal Carbide
   (b) Electron Sea Model in Metal bonding
   (c) Electron deficient compound
   (d) Nitrogen Fixation
   (e) Fullerene-C^{60}
   (f) Solid solution.

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